Aptamer-Functionalized Hydrogel Catheter Coating with Embedded Depolymerase to Prevent Catheter Associated Urinary Tract Infections

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Executive Summary

Catheter-associated urinary tract infections (CAUTI) are responsible for over **30% of all healthcare associated infections**. In nursing homes, CAUTIs are one of the most common infections and 50% of residents end up with symptomatic CAUTIs. Severe cases of CAUTI can have led to **100,000 hospitalizations per year** in the United States. The primary **cause of these CAUTIs is the *Proteus mirabilis,*** and this pathogen has developed antibiotic resistance to most antibiotic treatments. This has shifted industry focus to preventative measures to reduce CAUTI. The most common options for prevention currently include silver hydrogel coatings, antibiotic coatings, and human prevention protocols such as sterile insertion and early catheter removal. However, these treatments exhibit poor long term efficacy, increase antibiotic resistance, and do not possess adequate anti-biofilm properties. Therefore, **we aim to develop a CAUTI prevention system within a hydrogel coating for catheters to address multi-drug resistance of *P. mirabilis* and ultimately reduce CAUTI infection rates in hospitals and nursing home populations.**

P. mirabilis is a gram negative bacteria, which uses **quorum sensing** to adapt gene expression based on surrounding cell density; this enables bacteria to become more virulent and increase swarming mechanisms. This quorum sensing pathway begins with the binding of autoinducers, specifically **AHLs**, therefore by binding the autoinducers it is possible to prevent the signal proliferation and is also used as a signal for the presence of bacteria. Another method of preventing biofilm formation is through the use of bacteriophages. A tail spike protein of the **vB_PmiS_Pm-CJR bacteriophage** has been isolated and found responsible for degradation of the biofilm via polygalacturonic acid degradation and swarm inhibition. **Our design proposes a novel hydrogel coating that utilizes quorum quenching of AHLs via aptamers. These aptamers will act as part of the hydrogel crosslinking mechanism and when bound to AHLs, will break the crosslinks and release the depolymerase from the hydrogel.**

Our coating for CAUTI prevention must meet several design criteria for success: **high sensitivity, ensure biocompatibility and reduce coating side effects, effective at biofilm formation prevention, and have long term efficacy**. To ensure these design criteria are met **our testing plan involves six phases: Selection of aptamer and verification of depolymerase activity, in-vitro testing, ex-vivo testing, in-vivo testing, safety and biocompatibility testing, and clinical trials.** By utilizing a dual pronged approach for the prevention of *P. mirabilis* biofilm formation, our design promises an effective, safe, and reliable alternative to current prevention methods.

Relevant Biological Background

Catheter-associated urinary tract infections (CAUTIs) are responsible for more than 30% of all healthcare-associated infections [1]. Furthermore, approximately 75% of all UTIs developed in hospitals are due to the usage of a urinary catheter, and about 20% of hospitalized patients use urinary catheters [2]. While the direct mortality rate of CAUTI is fairly low, the infection leaves patients with a higher cumulative infection burden and is the second leading cause of secondary nosocomial bloodstream infections. Additionally, CAUTIs generate bacteremias that have an associated mortality of 10%[1]. In nursing homes CAUTIs are one of the most common infections, with 50% of catheterized residents developing CAUTIs with symptoms [3]. These CAUTI symptoms include but are not limited to acute change in mental status, fever, leukocytosis, and hypotension [4]. CAUTIs can also lead to more severe consequences, particularly in nursing homes where catheter usage is more long-term. Patients with CAUTIs can develop sepsis, pyelonephritis which accounts for 100,000 hospitalizations in the US per year [5], urethra stenosis, and encrustation [6]. Patients with long-term indwelling catheters are more likely to develop a UTI with multidrug resistant (MDR) organisms, and these MDRs can be transferred to other residents without long-term indwelling catheters [7].

While standard UTIs are typically caused by pathogens of the Enterobacteriaceae family, such as *Escherichia coli*, *Proteus mirabilis* is the specific pathogen that is directly related to the usage of urinary catheters [8]. *P. mirabilis* is a gram-negative facultative anaerobe that has the ability to secrete an adhesive polysaccharide and self-elongate. These traits enable this pathogen to easily attach to catheters and intravenous lines[9]. *P. mirabilis* has high virulence through a variety of mechanisms. *P. mirabilis* has been shown to trigger urease-induced stone formation [10], and this is a critical method of antibiotic resistance for the bacteria because they can become embedded in these stones which then protect the pathogen from the immune system and antibiotics [11]. Furthermore, *P. mirabilis* carries integrative and conjugative elements that enable self-replication and self-transfer to other strains, this enables the pathogen to transfer its virulence genes and antibiotic resistance [12]. *P. mirabilis* actively switches between a shortened flagella swimming state and a longer flagella swarmer state which is another cause for antibiotic resistance. Since the pathogen can create robust swarming patterns on solid surfaces, it has an improved biofilm formation [11].

Overall *P. mirabilis* has formed a resistance to the following antibiotic treatments: trimethoprim-sulfamethoxazole (SXT), Beta-lactams, fluoroquinolones, nitrofurantoin, fosfomycin, aminoglycosides, sulfonamides, and tetracyclines. In addition to these common resistances, 3 out of 35 strains of *P. mirabilis* are pan-drug resistant, which means that they are resistant to all 10 antimicrobial classes [13]. Quorum sensing offers a novel way to target pathogens that have developed antibiotic resistance. Quorum sensing is a process in gram-negative bacteria that enables bacteria to modify their gene expression in response to changes in cell densities and surrounding species compositions [14]. The primary signalling molecules utilized are N-acyl homoserine lactones (AHL), and these AHLs trigger LuxR family transcription. This particular family of genes heavily influence swarm cell differentiation, biofilm formation, and swarming patterns [15]. Additionally AHLs secreted by other gram-negative bacteria during polymicrobial UTIs also have been shown to trigger *P .mirabilis* virulence and shows that *P. mirabilis* is capable of interspecies signaling [15] [16].

Aptamers are single stranded DNA or RNA molecules that bind to protein targets isolated from systematic evolution of ligands by exponential enrichment(SELEX)[17]. In general, aptamer have high affinity for their ligand and very strong specificity. SELEX has been used to develop aptamers that bind to gram negative bacteria AHL's, which as aforementioned are the autoinducers to the quorum sensing pathway [18].

Additionally due to this multi-drug resistance, prevention and early degradation of *P. mirabilis* biofilm formation has been cited as a critical method of prevention of CAUTIs [11]. *Proteus* bacteriophage (vB_PmiS_Pm-CJR) is a novel bacteriophage found in the mud of parkland regions of Northern Ireland, and its DNA was sequenced and a region of the phage

was found to code for a pectate lyase domain of the tail spike protein that is responsible for biofilm degradation [19]. This tail spike protein from the *Proteus* bacteriophage is responsible for biofilm depolymerase in *P. Mirabilis* [19]. This tail spike protein works by targeting and degrading polygalacturonic acid [19]; Polygalacturonic acid is a key component of *P.mirabilis*'s outermembrane lipopolysaccharide and also facilitates in the migration of swarmer cells [20]. Therefore, the tail spike proteins degrade the biofilm via membrane degradation, and it also prevents biofilm formation by impending swarming.

Catheters are hydrophobic and have high surface friction, therefore many hydrogel coatings have been developed to improve the lubricity and comfort of patients [21]. The future of catheter hydrogel coatings emphasizes the antimicrobial properties of the coating. This usage of hydrogel coatings, multidrug resistances, and various virulent genes motivates our design proposal. We propose a novel hydrogel coating consisting of aptamer functionalized crosslinks, specific to AHL quorum sensing molecules, that when bound to AHL release the tail spike depolymerase protein from the hydrogel, causing degradation and prevention of *P. mirabilis's* biofilm.

Markets/Needs Assessment

Healthcare-associated infections (HAIs) represent a significant burden on healthcare systems globally, with CAUTIs accounting for approximately 30% of all HAIs. In the United States alone, there are an estimated 449,334 CAUTI cases annually, leading to significant healthcare costs and patient morbidity [22]. With approximately 20% of hospitalized patients requiring urinary catheterization and 75% of hospital-acquired UTIs being catheter-related, the market for CAUTI prevention technologies is substantial and growing [23].

Economically, the global urinary catheter market was valued at $4.18 billion in 2022 and is projected to reach $6.32 billion by 2030, growing at a CAGR of 5.3% [24]. The economic burden of CAUTIs extends beyond direct medical costs, with each CAUTI case estimated to cost healthcare facilities between $876 to $10,197 per incident [25]. Additionally, since 2008, the Centers for Medicare & Medicaid Services (CMS) no longer reimburse hospitals for costs associated with preventable CAUTIs, creating a strong financial incentive for healthcare facilities to invest in effective prevention strategies.

Current CAUTI prevention strategies are divided into three main categories. Antimicrobial-coated catheters such as silver-coated ($15-25 per unit) and antibiotic-impregnated catheters (e.g., nitrofurazone, minocycline; $20-30 per unit) are limited by a short efficacy duration (7–14 days) and the growing threat of antimicrobial resistance [26], [27]. Hydrogel-coated catheters ($10-15 per unit) improve patient comfort through reduced friction but provide limited infection prevention due to weak antimicrobial properties [28]. Lastly, prevention protocols focus on sterile insertion, early catheter removal, and bundled care approaches but are labor-intensive and dependent on consistent compliance [29].

Competing treatment options include innovations addressing the limitations of existing technologies. Current antimicrobial treatments face diminishing efficacy against MDR pathogens, while new biofilm inhibitors, such as bacteriophage-derived depolymerase proteins, are emerging as potential adjunctive therapies. Novel hydrogel coatings with embedded antimicrobial agents or aptamers targeting quorum sensing pathways are gaining attention for their precision against biofilm-forming pathogens like *P. mirabilis*. Our design proposal builds on these advances by incorporating aptamers specific to quorum sensing, coupled with bacteriophage tail spike proteins for targeted biofilm degradation. This approach promises to address the limitations of current prevention strategies, particularly against MDR biofilm-associated infections, offering a more effective, innovative solution to

CAUTI management. To evaluate its own market position and potential impact, we have conducted a comprehensive SWOT analysis:

Table 2: SWOT Overview of Proposed Aptamer-Functionalized Hydrogel Catheter Coating with Embedded Depolymerase

Strengths	Weaknesses	Opportunities	Threats
• Novel dual-action mechanism combining quorum sensing detection and biofilm degradation • Specific targeting of P. mirabilis through tailored aptamer design and phage-derived depolymerase • Biocompatible polyacrylamide hydrogel platform with proven safety profile • Smart-release mechanism triggered only when needed, reducing unnecessary antimicrobial exposure • Modular design allowing component optimization and potential adaptation for other pathogens • Extended functional lifetime compared to traditional antimicrobial coatings	• Complex multi-component system requiring precise optimization of: aptamer-AHL binding efficiency, depolymerase stability in hydrogel matrix, cross-linking density, and mesh size • Temperature and pH sensitivity of biological components • Higher initial production costs compared to standard catheters • Limited shelf-life of biological components	• Rising costs of CAUTI treatment driving innovation adoption • Potential application in other medical devices prone to biofilm formation • Partnership possibilities with established catheter manufacturers • Expansion to target other bacterial species through aptamer modification • Patent protection potential for novel biological-mechanical system • Growing market in developing countries with rising healthcare infrastructure	• Competition from emerging alternative technologies • Potential resistance development through bacterial evolution • Risk of enzymatic degradation of the aptamer or depolymerase components in the urinary environment • Variability in AHL concentration levels between patients affecting system reliability • Market competition from established catheter manufacturers developing their own smart-coating technologies

The global urinary catheter market segmentation shows particular growth in specialty catheters (CAGR 7.2%), with infection prevention devices commanding premium pricing. Our technology addresses the highest-value segment of this market, with potential for expanded applications in other medical devices prone to biofilm formation.

In conclusion, while current CAUTI prevention methods offer incremental improvements, our aptamer-depolymerase hydrogel technology represents a transformative approach that addresses fundamental limitations in existing solutions. The integration of phage-derived depolymerase with AHL-specific aptamers in a biocompatible polyacrylamide hydrogel matrix offers a sophisticated, targeted approach to preventing *P. mirabilis* biofilm formation. This smart-release platform directly addresses the $4.18 billion urinary catheter market's urgent need for effective, non-antibiotic prevention strategies, particularly given the rising costs of CAUTI treatment and increasing antimicrobial resistance. Our system's dual-action mechanism—combining early detection through quorum sensing molecules with targeted biofilm degradation—provides a distinct advantage over current single-approach prevention methods. While the technology faces challenges in optimization and regulatory approval, its potential to significantly reduce CAUTI incidence rates and associated

healthcare costs, coupled with its adaptability to other bacterial pathogens, positions it as a promising innovation in medical device coating technology.

Design Criteria

Four main design criteria make our aptamer-functionalized hydrogel catheter coating a safe, reliable, and competitive catheter option in the current market. One of the most critical design criteria surrounds the **sensitivity of AHL binding** to our aptamer. This is critical to ensure that the quorum-sensing pathway is both sensed and interrupted. The next critical design criteria is **effective biofilm prevention**. This is the main determinant of whether our catheter coating prevents CAUT infections. The next two design criteria ensure that our method is competitive in the current market. The **safety** of our product is critical, this coating must be biocompatible and our method of biofilm prevention should not cause unintended side-effects. The next criteria surrounds the **long-term efficacy** of the system. Our design criteria, as shown in Table 2, focuses on resolving common concerns with existing treatment options.

Table 3: Summary of design criteria in proposed hydrogel coating for CAUTI Prevention

Design Criteria	**Proposed Primary Design (PeaS) Goals**	**Antimicrobial-coated catheters**	**Silver Hydrogel coated Catheters**	**Prevention Protocols: Sterile insertion, early catheter removal, bundled care**
Sensitivity	AHL to Aptamer Binding Affinity **10pM<Kd<10nM**	Not Triggered, sustained slow release	Not Triggered, sustained slow release	Based on Human Precision and Practice
Safety: Coating Side Effects and Biocompatibility	→ minimize disruption of “good” bacteria → reduce surface roughness	→ increased surface roughness → increased bacterial drug resistance [30]	→ silver ion leakage → hepatotoxicity → argyria [23]	→ no significant side effects
Effective Biofilm Prevention	Decrease CAUTI by **over 85%**	Decreased CAUTI by **65%** [22]	Decrease CAUTI by **21%** [31]	Decrease CAUTI by **53%**[32]
Long Term Efficacy	Protect Patients from CATUI for **more than 4 weeks**	Protect Patients from CATUI for **1-2 weeks** [26]	Protect Patients from CATUI for **1-2 weeks**[26]	-No sustained release mechanism, effective as long as in practice

Long-term Efficacy

One of the major concerns for the treatment and prevention of CAUTIs for patients is the long-term efficacy of the method. For long-term patients (catheterized for more than a month) catheters are changed every 4 to 12 weeks and for short-term patients(catheterized for less than a month) catheters are changed every 2 to 3 weeks [4] [20]. Additionally, there are several barriers in place for timely catheter removal. An observational and interview-based study found that physicians do not routinely review catheter necessity, use ‘do not remove’ orders, have difficulty finding catheter data, and have communication barriers with fellow clinicians in regard to catheter usage [34]. This data is further supported in other studies that found that in US hospitals less than 25% of hospitals had a monitoring system for catheters and less than 20% monitored the duration and discontinuation of

catheters [35]. Given these timelines and the difficulty in adherence to these timelines it is critical that our coating prevents *P.mirabilis* growth for a competitive amount of time. The effectiveness of antibiotic-coated catheters and silver-impregnated catchers can prevent CAUTI growth in patients for 1-2 weeks [26], after this period of time the effectiveness of the antibacterial coating decreases. To be competitive with the market and to keep up with the changing patterns of both short and long-term catheters, our coating needs to prevent *P.mirabilis* growth for more than 4 weeks. This means that our coating must be loaded accordingly. One of the important features of aptamers is their reusability, after aptamers bind to these AHLs they can be regenerated without having the aptamers affinity compromised[36]. The next components of this long-term efficacy concern the properties of the hydrogel. Studies have shown that to prevent shedding of the hydrogel coating during catheter adhesion, the hydrogel must have an adhesion strength of approximately 40kPa [37]. This means that our hydrogel must have a similar or higher surface adhesion to the catheter. The next concern of the hydrogel is its degradation, for our hydrogel to continue binding AHLs it cannot degrade before this four-week mark. This helps motivate our choice of Polyacrylamide hydrogels. These hydrogels are known to have slow degradation *in vivo* and can remain stable in the body for up to several years [38]. This long-term stability is a reason why Polyacrylamide has been chosen for implantable applications and will ensure that our coating will be present and attached to the catheter for more than four weeks.

Sensitivity: AHL Binding to Aptamer

The next design criteria is critical to ensuring that our aptamer has a sufficiently high affinity for *P. mirabilis'* autoinducers, AHL. This extends the mechanism of the options currently on the market. Silver coated foley catheters work by having a slow sustained release of silver ions from the hydrogel, and this slow steady release ensures that bacterial adherence is lowered across the entire surface[39]. So the sensitivity of this coating is more based on the amount of silver ions released at a time to ensure a sufficient surface coating to prevent adherence. Moving into the sensitivity of antibiotic coated catheters, similar to silver coating, the sensitivity is based on controlled release of the antimicrobial agent in the hydrogel. However, since our design relies on aptamers to trigger the release of the depolymerase and it is not a slow sustained release, like the market alternatives, it is imperative that *P. mirabilis* AHLs have a high binding affinity to our selected aptamer. Typical aptamers are 10-100 times smaller than antibodies so we can arrange them with a higher density in the hydrogel [40], this is important because it helps improve the sensitivity of our hydrogel coating. Previous literature has determined the effectiveness of AHL binding to specific aptamers and has found strong affinity between AHL's and aptamers. To ensure the effectiveness of our design, and to compete with the sustain release method of alternative designs, our aptamers Kd must be measured [41]. Previous studies have shown that strong aptamer bindings have a Kd of between 10pM-10nM for small molecules like AHL's [42], and therefore our design must be on the lower end of this range for the strongest affinity.

Safety: Side Effects

Side effects of antibiotic catheters have been studied and reviewed. In 6 out of 13 trials adverse effects occurred. This included intense pain and hematuria, burning sensations, bleeding from catheter placement, and sepsis [43]. However, most of these side effects are due to catheter placement and handling. One of the concerns for antimicrobial coating is that it can increase surface roughness and this has been a cause of more discomfort upon catheter removal [30]. Another unintended side effect from antimicrobial

catheters includes an increase in bacterial drug resistance, which would enable *P. mirabilis* to continue to develop more virulent strains and increase MDR [31]. With regard to silver coated catheters, these can have silver ion leakages that produce local side effects [31]. This phenomenon has been better studied in silver containing dressings. Patients that had been treated with silver dressings had reported hepatotoxicity and argyria like symptoms, and researchers concluded that silver dressing wound treatment should not occur for longer than 30 days [44]. The primary concern of unintended side effects in regards to quorum quenching is the disruption of healthy microbiomes and the potential for microbes to develop resistance against quorum quenching therapies [45]. Bacteria such as Lactobacilli is the main uropathogen present in the bladder, and this bacteria has been considered to prevent UTIs, specifically in women [46]. Lactobacilli have also been shown to display their own quorum quenching mechanisms, so the disruption of this bacteria would reduce the body's natural ability to prevent UTIs [47]. These side effects partially encompass the biocompatibility of the hydrogel coating. To help ensure biocompatibility of our hydrogel, we will utilize prior usage of Polyacrylamide in biological application to ensure that it will be well received by the body [48].

Effective Biofilm Prevention

Prevention of biofilm formation is critical to ensuring patients are protected from CAUITs. For competitive treatments, silver-coated catheters have been shown to significantly reduce bacterial counts in common pathogens like *Staph*, *E. coli* and most importantly *P.mirabilis*. Specifically, the log count of gram-negative bacteria decreased between 5.13-6.13 logs [49]. However these silver coated catheters only reduced CAUTI in patients by 22%[31]. Another common antibiotic used to coat catheters includes nitrofurazone. Nitrofurazone can decrease CAUTI incidence rates by approximately 65%[21]. Additionally proper handling methods have only been shown to reduce CAUTI incidences by 53%[32], [51]. Other antibiotic coated catheters have only decreased CAUTI from 8.8% of patients to 8.3% of patients and this is not a significant decrease. Furthermore, while nitrofurazone has the best market option to prevent biofilm formation, *P.mirabilis's* swarming mechanism enables growing antibiotic resistance. By using the quorum sensing molecules as a method for quorum quenching and as a release method for depolymerase, our design seeks to reduce bacterial growth rates and improve prevention. Previous studies have found that quorum quenching in *P. mirabilis* can inhibit biofilm formation by up to 76%. Furthermore this quenching increased downregulation of virulent genes from 2.6 fold to 20.7 fold [52]. In research, the adhesion of *P. mirabilis* reduces bacterial adhesion by 10 fold and showed a significant decrease in colony counts [19]. The goal of combining both quorum quenching and utilizing a depolymerase will leverage both methods for biofilm inhibition and ultimately prevent CAUTIs.

Design Plan

Phage-encoded depolymerases are specialized enzymes derived from bacteriophages that have the ability to degrade the extracellular polymeric substances forming bacterial biofilms [53]. Thus, these enzymes present a potent biofilm disruption strategy, especially significant in medical device-associated infections.

Our proposed design aims to develop a hydrogel system functionalized with a specific aptamer that responds to the AHL quorum sensing molecule, commonly associated with bacterial communication and biofilm formation. This responsive system is intended for coating medical catheters to prevent CAUTI through a dual-faceted approach that combines quorum quenching and biofilm depolymerization.

Construction and Production of the Depolymerase:

For effective biofilm degradation, we will utilize a researched depolymerase, specifically the tail spike protein of the Proteus bacteriophage (vB_PmiS_Pm-CJR). This enzyme specifically targets polygalacturonic acid in *P. mirabilis* biofilms [19]. The protein sequence of the depolymerase is obtained from UniProt [54] and optimized using IDT for bacterial codon usage [55]. This gene encoding the depolymerase will be synthesized and cloned via Gibson Assembly into an expression vector (Addgene #105745) [56] tailored for high-level production in E. coli, as seen in Figure 2.

After transformation into E. coli, the expression of the depolymerase will be IPTG induced under controlled conditions to maximize yield. The enzyme will be purified using affinity chromatography, facilitated by a histidine tag added upstream of the depolymerase sequence to ensure purity and activity. The integrity and functionality of the purified enzyme will be confirmed through SDS-PAGE and an activity assay against polygalacturonic acid, specifically the dinitrosalicylic acid (DNS) assay which involves measuring the release of reducing sugars to quantify enzymatic degradation.

Figure 2: Plasmid map of vB_PmiS_PM-CJR Tail Spike Protein (Depolymerase) (4837 bp), highlighting key features including the HisTag, T7 promoter, RBS, lacI and Amp resistance (Kan-R) genes, and multiple restriction enzyme sites.
Benchling Link:

Integration of the Depolymerase into the Hydrogel

Once validated, the purified depolymerase will be incorporated into the hydrogel matrix during the gel formulation process. This integration will be designed to ensure that the enzyme is evenly distributed throughout the hydrogel, providing uniform biofilm degradation capabilities upon activation by the AHL-responsive aptamer system.

Design and Selection of the Aptamer:

The design of the DNA aptamer component for our system will begin with a known aptamer that specifically targets 3O–C12–HSL, a type of acyl-homoserine lactone (AHL) involved in quorum sensing in gram-negative bacteria such as *Pseudomonas aeruginosa* [18]. This known aptamer serves as a starting point due to its proven efficacy in binding AHL and modulating its activity, as demonstrated in model biosensor systems [18].

To enhance the specificity and binding efficiency of the aptamer towards AHL molecules produced by *P. mirabilis*, we will create a library of mutated aptamers based on the known sequence. This library will be generated through an error-prone PCR process, introducing random mutations to diversify the aptamer sequences. This method will allow us to explore a broad range of sequence variations and identify those with potentially higher affinity or specificity for the target AHL molecules.

Once the library is constructed, it will be subjected to the Systematic Evolution of Ligands by Exponential Enrichment (SELEX) process. This iterative technique will involve exposing the library to *P. mirabilis* purified AHL molecules. Aptamers that successfully bind to the AHL will be separated and amplified, while those that do not bind will be washed away. Each round of SELEX will refine the pool of aptamers, increasingly enriching it with sequences that have the highest affinity for the target molecule.

Integration of Aptamer into Hydrogel Crosslinks

Once the top aptamers are selected, the aptamers will be synthesized and attached to a custom-designed linker that aids in the formation of cross-links within the hydrogel matrix. This approach allows the aptamers not only to stabilize the hydrogel structure through cross-linking but also to prepare the system for a responsive action upon interaction with AHL molecules.

The mechanism of crosslink formation and breaking, adapted from the LinkerAdap system [57], is demonstrated in Figure 3. Here the linker sequence is shown in green and the aptamer in red. The aptamer-linker will hybridize with two engineered complementary DNA strands (black) conjugated to and embedded in the linear polyacrylamide chains (blue) of the hydrogel. The hybridization of aptamer-linker to both the DNA strands forms the polymer crosslinks.

In the presence of AHL quorum sensing molecules these crosslinks break. Upon AHL binding, the aptamer will undergo a conformational change due to competitive binding, thus, reducing its affinity for the complementary DNA strand (Strand 2 in Figure 3). This, in turn, leads to the controlled dissolution of the hydrogel matrix.

Figure 3: Graphical representation showing mechanism of formation and breaking of the aptamer functionalized crosslinks and the subsequent dissolution of the hydrogel and release of depolymerase

Mechanism of Action

As seen in Figure 3 in the presence of AHL molecules, indicative of *P. mirabilis* activity and biofilm formation, the cross-linking breaks, leading to the gradual dissolution of the hydrogel matrix. As the hydrogel breaks down, the embedded depolymerase enzymes are released into the surrounding environment where they actively prevent and degrade the biofilm formation. This targeted release mechanism ensures that the therapeutic action of the depolymerase is localized to the site of infection, enhancing the effectiveness of the treatment and reducing the likelihood of bacterial resistance development. Simultaneously, the aptamer's interaction with AHL disrupts the quorum sensing pathway, further inhibiting biofilm maturation and bacterial communication, thus providing a dual approach to combating CAUTIs.

Hydrogel Formulation

We will utilize a biocompatible hydrogel as the matrix to embed the depolymerase and crosslink with the aptamers. As per previous research, polyacrylamide-based hydrogel (PAHG) coatings improve biocompatibility of implanted devices [58]and PAHG has also been used as the matrix for responsive hydrogels [59]. Referencing established literature [57], we will create two sets of acrydite-modified DNA oligonucleotides (Strands 1 and 2 in Figure 3), which will then each be copolymerized with acrylamide. This allows for the integration of the DNA strands into the polymer chains of the hydrogel. These acrydite-modified DNA have demonstrated similar functioning and characteristics to acrylamide monomers and can thus be incorporated efficiently in the polymer hydrogel. We then form a fluid solution by combining the two DNA-polyacrylamide solutions in stoichiometric ratios. This fluid then transitions into a gel upon the introduction of the specially designed cross-linking aptamer-linker oligonucleotides.

Alternative Design Plans

An alternative approach, as seen in Figure 4, to our primary design focuses on a simplified yet potentially more versatile quorum sensing inhibition strategy. Instead of incorporating the depolymerase release mechanism, this design utilizes multiple specialized aptamers that directly target and sequester various quorum sensing molecules, including AHL, AI2, and PQS, preventing them from binding to their native receptors in infecting bacteria.

By removing the depolymerase component, we eliminate potential challenges associated with protein stability in the hydrogel matrix and reduce the complexity of the system's manufacturing process. The design would incorporate a panel of different aptamers, each optimized through SELEX to target distinct molecules, providing a more holistic CAUTI prevention mechanism. These aptamers would be directly integrated into the same PAHG matrix, but without the need for complex release mechanisms. When these molecules are produced and released by colonizing bacteria, the aptamers would act as molecular sponges, effectively disrupting bacterial communication before biofilm formation can be initiated.

This preventative approach, while potentially less aggressive than the depolymerase-based system, offers advantages in terms of broad spectrum targeting as well as manufacturing simplicity and storage stability by eliminating the biological complexity of protein-based components.

Figure 4: Graphical representation showing mechanism of quorum quenching by aptamers

Testing and Implementation Plan

Our testing plan for the primary design will be divided into 6 key phases which involve **selection of aptamers and verification of depolymerase activity, in-vitro testing, ex-vivo testing, in-vivo testing, safety and biocompatibility testing** and finally **clinical trials**.

Phase 1: Selection of Aptamer and verification of depolymerase anti-biofilm activity

Firstly, the binding affinity of the aptamers for AHL will be investigated using Surface Plasmon Resonance (SPR), which provides quantitative metrics such as the dissociation constant (K_d), to determine the most suitable aptamer for use in the hydrogel's application. To ensure specificity of the aptamer, counterselection against other bacterial quorum sensing molecules will also be conducted. Secondly, in order to ensure that the selected depolymerase derived from the Proteus bacteriophage effectively degrades biofilms caused by *P. mirabilis*, a known concentration of the purified depolymerase will be incubated with biofilms of the target bacteria and a Minimum Biofilm Inhibitory Concentration (MBIC) assay [60] will also be conducted to determine the lowest concentration of the depolymerase which is required to prevent the initial formation of the biofilm under typical physiological conditions. The MBIC assay will inform appropriate dosing of the depolymerase in the hydrogel.

Figure 5: Example of MBIC assay workflow [61]

In both instances, crystal violet staining [62] will be utilized to investigate the anti-biofilm activity of the depolymerase. The crystal violet staining procedure works by staining the biomass of bacterial biofilm with a dye. The absorbance of the dye is measured after it is bound to the biofilm matrix, thus facilitating the quantification of biofilm degradation after exposure to an anti-biofilm agent. A decrease in absorbance correlates to a reduction in biofilm formation which indicates efficacy of the depolymerase. Negative control experiments without depolymerases will also be included to allow for comparison.

Phase 2: In-vitro Testing

Once the individual components of the hydrogel have been verified, in-vitro testing of the hydrogel coating can be conducted. The in-vitro testing strategy involves investigating the ability of the coating itself to prevent *P. mirabilis* biofilm formation. In this experiment, the Calgary Biofilm Device (CBD) [63] can be leveraged for high-throughput analysis of the hydrogel coating's efficacy. Essentially, this device will allow for 96 wells to be inoculated with a concentration of *P. mirabilis* which facilitates the formation of biofilms on the surface of the wells. Subsequently, four treatment groups of hydrogels will be investigated. These are: Hydrogels with aptamer-depolymerase, Hydrogels with depolymerase only, Hydrogels without aptamer or depolymerase (control) and finally untreated bacterial culture (negative control). These hydrogels will be coated on the pegs of the device to facilitate incubation with the bacteria and each treatment condition will be performed in biological triplicates to ensure validity of the results obtained. Once again, crystal violet staining will be used to determine the degree of anti-biofilm activity observed in each treatment condition over 24, 48 and 72 hour time periods. By conducting observations at regular time intervals, insight into the potential longevity of the system will also be provided. The results of the crystal violet staining can also be verified using live/dead staining and confocal microscopy in conjunction with an open source analysis tool such as Biofilm Viability Checker [64]. In addition, this in-vitro experiment will be conducted using non-biofilm forming bacteria to investigate the off-target effects of the hydrogel.

Phase 3: Ex-Vivo Testing

Upon successful in-vitro testing, ex-vivo testing can be conducted. Ideally, this aspect of the testing plan involves coating catheters with the hydrogel and investigating the anti-biofilm activity under conditions which mimic urinary catheters. This can be achieved by using a flow chamber which simulates urine/blood conditions along with bacterial suspensions around the catheter which would aid in replicating real-life biofilm formation. Both coated and uncoated (negative control) catheters will be tested under these conditions and anti-biofilm activity will be determined quantitatively using crystal violet staining and confocal microscopy. Confocal microscopy would provide detailed images of the biofilm structure and thickness, allowing visualization of potential biofilm formation on the catheter. In the absence of a flow chamber, the coated and uncoated catheters can simply be submerged in simulated body fluid and similar analysis conducted.

Phase 4: In-Vivo testing

For In-vivo testing, pigs may be a suitable animal model due to the similarity of their urinary tract to humans. In-vivo testing will be carried out by first inducing conditions of catheter associated urinary tract infections in the animal model. This can be achieved by inoculating a controlled dose of bacteria (*P. mirabilis*) in the bladder. Subsequently, hydrogel coated catheters as well as uncoated catheters will be inserted into the bladder through the urethra. Over a 7 day period, the urine will be tested to measure bacterial load and at the end of 7 days the catheters will be removed and the formation of biofilms on the surface of the catheter will be investigated.

Phase 5: Safety and Biocompatibility

The hydrogels will need to be tested for their safety and biocompatibility. Firstly, cytotoxicity testing using an MTT assay [65] will be carried out to ensure that the hydrogel is non-toxic to human cells. The hydrogel's compatibility with blood will also be investigated via

a hemocompatibility assay [66]. Finally, immunogenicity of the hydrogel will be investigated via a cytokine release assay which measures the release of pro-inflammatory cytokines from immune cells exposed to the hydrogel.

Phase 6: Clinical trials

Finally, after all previous criteria have been met and our hydrogel has been refined, clinical trials will be conducted. Throughout the clinical trial, the health of patients will be consistently monitored to ensure unwanted or off-target effects are not experienced. Ultimately, the key metrics investigated will be the anti-biofilm activity of the hydrogel coating as well as any signs of infection in the patient.

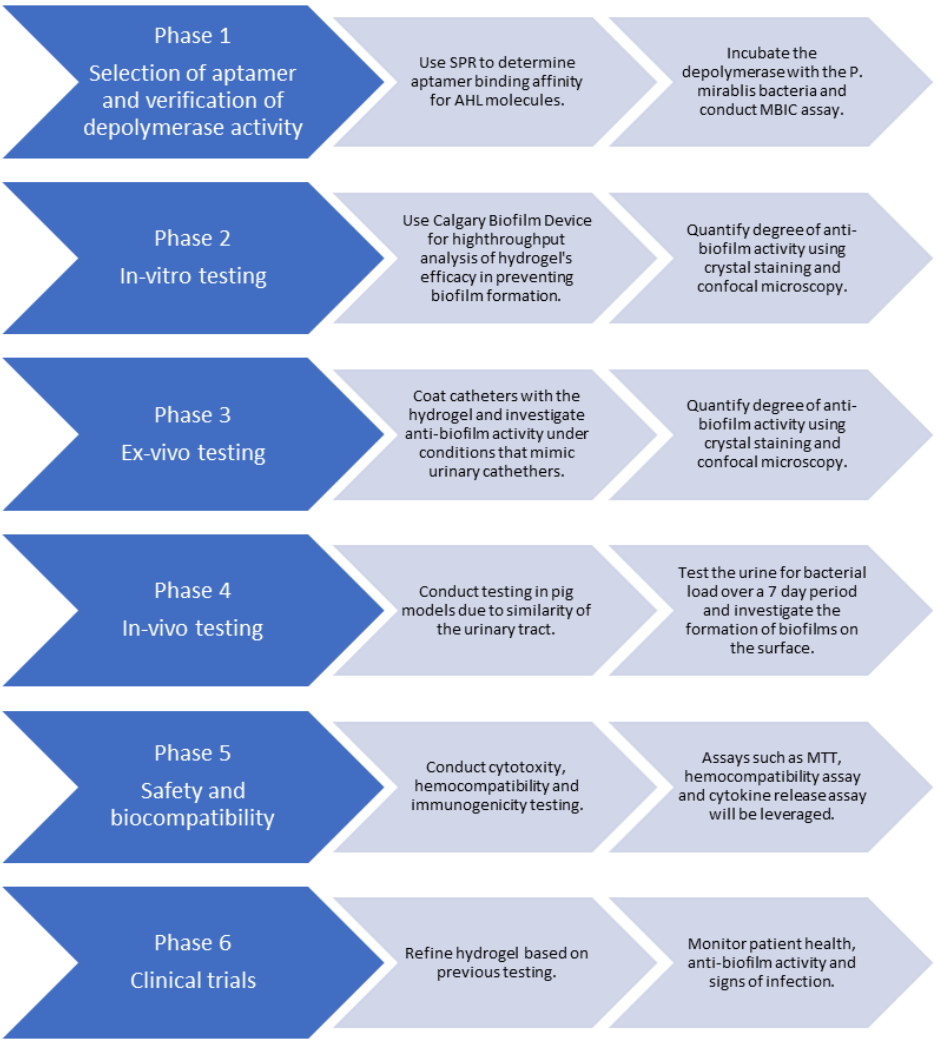


Figure 6: Flowchart Detailing Testing and Implementation Plan for the Aptamer-Functionalized Hydrogel with Depolymerase System

Regulatory Analysis

In this project, we are developing an aptamer-functionalized hydrogel-coated catheter embedded with depolymerases to target CAUTI-causing bacteria and their quorum-sensing molecules. The product will be classified as a combination product [67], which integrates a medical device (urinary catheter with hydrogel coating) and biological components (aptamers and depolymerase). In this case, The FDA's Office of Combination Products (OCP) will assign a lead center, which is likely to be Center for Drug Evaluation and Research (CDER) given the therapeutic action of the aptamer and the depolymerase. Meanwhile, urinary catheters are generally classified as Class II medical devices under 21 CFR 876.5130, and the biological component of the design may elevate the risk profile, leading to a Class III classification, which requires Premarket Approval (PMA). As the primary goal is to prevent CAUTIs using the aptamer-functionalized hydrogel-coated catheter with depolymerases, the product requires rigorous evidence demonstrating the capacity of preventing UTIs compared to uncoated catheters. To prepare for the Investigational New Drug (IND) application, we will conduct comprehensive preclinical studies of the product. These studies will include:

1. In Vitro Assessments:

Evaluate the hydrogel's ability to inhibit biofilm formation and quorum-sensing activity in the presence of *P. mirabilis.* Test stability and release profiles of the depolymerases under simulated urinary conditions.

2. In Vivo Animal Studies:

Rodents UTI model: Used for initial safety and efficacy testing, as smaller animals facilitate detailed characterization of the product.

Pig model: Selected for catheter insertion and functional studies due to their anatomical similarities to humans, providing more relevant data for clinical application.

The preclinical studies are expected to be completed within 10 months. These studies aim to demonstrate a reduction in bacterial colonization, the absence of biofilm formation, and no tissue irritation or adverse effects. Upon completion, the preclinical data will be compiled along with manufacturing details and clinical trial protocols for submission of the IND application.

Following the 30-day FDA review period for the IND application, the product will proceed to clinical trials.

In Phase 1, the objective will be to assess the safety, tolerability, and pharmacokinetics of the device. We plan to enroll 80 volunteers requiring catheterization for this study, which is expected to last approximately six months. We anticipate no adverse events during this phase.

In Phase 2, we will focus on evaluating efficacy and further assessing safety. A total of 300 patients at high risk of catheter-associated urinary tract infections (CAUTIs) will be recruited and divided into experimental and control groups. This phase is expected to span 1–2 years, during which we aim to observe a reduction in UTI incidence and the absence of biofilm-related infections.

For Phase 3, approximately 2,000 patients requiring catheters will be enrolled to confirm efficacy and long-term safety across a larger, more diverse population. This phase is anticipated to take 2–4 years. The primary endpoints include a statistically significant reduction in UTI rates, long-term safety, and biocompatibility of the device.

This phased approach ensures a rigorous evaluation of the device's safety and efficacy, supporting its potential to address the challenges of effectively preventing CAUTIs. The data

from preclinical assessment and clinical trials would then be submitted to FDA for premarket approval (PMA), and the review time typically takes 6-10 months.
In conclusion, the process for bringing the design to market is complex and is expected to take approximately 5 to 10 years.

IP Patent Landscape Overview

Field: Aptamer and Depolymerase Embedded Hydrogel Coating to Prevent CAUTI
Search Terms: “Hydrogel-coated catheters”, “Quorum-quenching aptamer”, “Biofilm depolymerase”, “Biofilm prevention technology”, “Catheter-associated urinary tract infections”, “Hydrogel-coated catheters AND biofilm prevention”
Patent Databases: USPTO, Google Patents, Espacenet
Results: USPTO: “Hydrogel coated catheters” → 815 Hits “Quorum-quenching aptamer” → 50 Hits “Biofilm depolymerase” → 229 Hits “Biofilm prevention technology” → 10881 Hits “Catheter-associated urinary tract infections” → 1273 Hits Google Patents: “Hydrogel coated catheters” → 2906 Hits “Quorum-quenching aptamer” → 20 Hits “Biofilm depolymerase” → 17 Hits “Biofilm prevention technology” → 134 Hits “Catheter-associated urinary tract infections” → 2666 Hits Espacenet: “Hydrogel coated catheters” →7495 Hits “Quorum-quenching aptamer” → 11 Hits “Biofilm depolymerase” → 122 Hits “Biofilm prevention technology” → 11080 Hits “Catheter-associated urinary tract infections” → 934 Hits
Key Patents Patent US8911424B2 [68]: This patent describes a method and device for preventing urinary tract infections in catheterized patients. The device is a cylindrically shaped antimicrobial hydrogel that can be applied directly over a inserted urinary catheter. The device leverages the antimicrobial activity of silver iodide to prevent bacterial infection. Patent US9259513B2 [69]: This patent describes a photocatalytic method to activate a reaction on the catheter surface that generates oxidizing agents in the form of reactive oxygen species (“ROS”) to destroy microorganisms in a biofilm that is present or forming. Patent CN117899273A [70]: This patent discloses an invention of a medical catheter hydrogel coating capable of firmly lubricating and resisting bacteria. The invention applies an antibacterial monomer composed of zinc dimethacrylate, zinc acrylate, and zinc monomethacrylate to resist bacterial infection.

Claim of Novelty:

Our innovation is an aptamer-functionalized hydrogel-coated catheter designed to prevent and treat catheter-associated urinary tract infections (CAUTIs) by embedding depolymerases directly into the hydrogel matrix. The aptamers selectively bind quorum-sensing molecules to disrupt bacterial communication, while the depolymerases degrade biofilms and eliminate bacteria. This dual-action approach uniquely integrates quorum-sensing inhibition and biofilm degradation,providing a biocompatible, targeted, and effective solution to reduce infection rates and combat multidrug-resistant pathogens.

Stakeholder Analysis and Ethical Considerations

Our project introduces a hydrogel-coated catheter embedded with aptamers and biofilm depolymerases to prevent and treat catheter-associated urinary tract infections (CAUTIs). The following analysis explores stakeholders' influence, expectations, and strategies needed to engage them throughout the development and implementation process. By mapping stakeholders in a relatively decreasing order of influence, we aim to ensure the product is safe, effective, and accessible while addressing ethical concerns related to its use.

Relevant Stakeholders

1. Patients: The primary beneficiaries of this innovation, particularly patients requiring long-term catheterization or those at high risk for CAUTIs. While they have limited direct influence, their experiences with current catheter technologies and concerns about infection prevention will drive demand. Ensuring the product is comfortable, accessible, and effective will enhance its acceptance. Patient feedback on usability and outcomes will shape iterative improvements.

2. Regulatory Agencies: Organizations such as the FDA and international counterparts will have large influence as they determine whether the product can be legally approved for use. Their requirements for preclinical and clinical validation will shape the development process. Regular engagement with these agencies is critical to ensure compliance with safety and efficacy standards for combination products. Adherence to regulatory guidelines is essential for commercialization.

3. Healthcare Providers: Key stakeholders include urologists, infectious disease specialists, and hospital procurement teams. Their willingness to adopt the catheter in clinical settings will significantly impact success. They have high influence and high interest as gatekeepers to patient care. Engaging them through clinical trials, training, and evidence-based data demonstrating safety and efficacy will be critical for product adoption.

4. Clinical Laboratories and Hospitals: Laboratories conducting quality control and hospitals implementing the device are crucial for ensuring the product’s proper integration into clinical workflows. They have medium to high influence in terms of operational success. Their feedback on technical and logistical challenges, as well as cost considerations, will inform the design and deployment strategies.

5. Research Institutions and Manufacturers: These stakeholders are critical in refining the design, scaling production, and ensuring quality control. Manufacturers have substantial influence over the affordability and reliability of the product. Collaboration with research institutions will drive innovation, while manufacturers’ decisions on pricing and distribution will affect market access.

6. Health Insurance Companies: Insurance providers will play a crucial role in determining

coverage and reimbursement policies, impacting patient access and market success. They have high influence and moderate interest in cost-effective solutions for reducing CAUTI-associated healthcare costs. Early engagement to demonstrate the economic benefits of the product will be key to securing coverage.

7. Hospital Infection Control Committees: These committees evaluate and approve new products for infection prevention within healthcare settings. They have moderate to high influence over product adoption. Their interest in reducing hospital-acquired infections aligns with the goals of the hydrogel-coated catheter. Providing data on infection rate reductions and cost-effectiveness will support their decision-making.

8. Advocacy and Patient Support Groups: Organizations focused on improving healthcare for patients with chronic conditions or those requiring catheterization can influence public opinion and healthcare policy. These groups have moderate influence and high interest in promoting innovative solutions for infection prevention. Their support can drive public awareness and funding opportunities.

9. Investors and Funding Agencies: These stakeholders provide necessary capital for research, development, and commercialization. They have high influence over project timelines, scale, and market entry strategies. Transparent communication about milestones and potential returns will ensure continued support.

Ethical Considerations

The development and implementation of our aptamer-functionalized hydrogel-coated catheter embedded with depolymerases to prevent and treat CAUTIs will adhere to the four pillars of bioengineering ethics: autonomy, justice, nonmaleficence, and beneficence.

1. Informed Consent and Transparency: Patients and healthcare providers must clearly understand the purpose, benefits, and limitations of using the hydrogel-coated catheter, especially concerning its role in preventing CAUTIs.

Resolution: Develop educational materials for healthcare providers and patients that explain the mechanism, expected outcomes, and potential risks, ensuring the information is clear and understandable for non-experts. Encourage open communication to address any concerns.

2. Risk of Adverse Reactions: The use of embedded aptamers and depolymerases may carry risks of adverse effects, such as local irritation or unforeseen immune responses.

Resolution: Conduct extensive preclinical and clinical testing to identify and mitigate any potential adverse effects. Ensure robust post-market surveillance to monitor long-term safety in diverse patient populations.

3. Risk of Ineffectiveness: If the hydrogel fails to prevent biofilm formation or bacterial colonization, patients may still develop CAUTIs, leading to potential harm.

Resolution: Rigorously validate the efficacy of the product through in vitro and in vivo testing and clinical trials. Clearly communicate the importance of continued infection monitoring alongside the catheter's use.

4. Equitable Access and Potential Health Disparities: High costs or limited availability may restrict access to underserved or low-resource populations, exacerbating healthcare disparities.

Resolution: Explore tiered pricing models, partnerships with insurance providers, and initiatives to supply the device to underserved communities at reduced costs. Advocate for coverage under public and private health insurance schemes.

5. Privacy and Confidentiality of Data: Clinical data generated during the device's use, such as infection rates or patient outcomes, must be securely managed to prevent

unauthorized access or misuse.

Resolution: Implement stringent data protection measures, including encryption and restricted access protocols. Inform patients and healthcare providers about how data will be used, stored, and protected, and obtain explicit consent for any data collection.

6. Impact on Patient Autonomy: Patients may feel obligated to use the catheter due to recommendations from healthcare providers, potentially leading to pressure or loss of autonomy.

Resolution: Provide balanced and transparent information about the device's benefits and limitations. Empower patients to make informed decisions by involving them in discussions about their care options.

7. Ethical Marketing and Communication: Overstating the device's capabilities or downplaying its limitations in marketing materials could mislead healthcare providers and patients.

Resolution: Ensure that all marketing and educational materials present accurate, evidence-based information about the device's effectiveness, safety, and proper use.

8. Cultural and Social Implications: Attitudes toward medical devices and infection prevention strategies may vary across cultures, potentially affecting acceptance and adoption.

Resolution: Develop culturally sensitive educational programs for both healthcare providers and patients. Adapt informed consent processes and marketing strategies to respect cultural norms and values, ensuring the device is accessible and acceptable in diverse healthcare settings.

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